[1 mark for correct forces; 1 mark for identification of forces]

Q3(a)

The heat supplied is used to increase the intermolecular potential energy and also to do work against the atmosphere as the vapour expands. [1]

The molecular KE remains constant on average , so there is no change in temperature. [1]

Q3(b)(i)

At A, $pV = \mu RT$, where μ = no. of moles of gas.

$(8.0 \times 10^5)(2.4 \times 10^{-5}) = \mu \times 8.31 \times 526$

$\mu = 4.39 \times 10^{-3}$ mol [1]

$= 4.4 \times 10^{-3}$ mol (shown)

Q3(b)(ii)

At C, $pV = \mu RT$

$(0.80 \times 10^5)(12.0 \times 10^{-5}) = 0.0044 \times 8.31 \times T_C$

$T_C = 262.6$ [1]

$= 263$ K (shown)

Q3(b)(iii)

Apply 1st Law of Thermodynamics to process AB:

$\Delta U = Q + W$

$\Delta U = 0$ as process AB is isothermal. [1]

$0 = Q_{AB} + [-\mu RT_A \ln(V_B / V_A)]$

$Q_{AB} = \mu RT_A \ln(V_B / V_A)$

$= 0.0044 \times 8.31 \times 526 \ln(12.0 / 2.4)$

$Q_{AB} = 30.95$ J = **31.0 J (heat absorbed)** [1]

Q3(b)(iv)

Apply 1st Law of Thermodynamics to process BC:

$\Delta U_{BC} = Q_{BC} + W_{BC}$

As the volume remains constant, $W_{BC} = 0$.

$\Delta U_{BC} = Q_{BC} + 0$

$U_C - U_B = Q_{BC}$ [1]

$(3/2)\,\mu RT_C - (3/2)\,\mu RT_B = Q_{BC}$

$Q_{BC} = (3/2)\,\mu R\,(T_C - T_B)$

$Q_{BC} = (3/2) \times 0.0044 \times 8.31\,(263 - 526)$

$Q_{BC} = -14.4$ J

Amount of heat lost = **14.4 J** [1]

Q3(b)(v)

Work done on gas during process CD = $W_{CD} = \mu RT_C \ln(V_D / V_C)$

But $T_C = (1/2)\,T_A$

$W_{CD} = \mu R\,(1/2)T_A \ln(V_A / V_B)$

$W_{CD} = (-1/2)\,\mu R\,T_A \ln(V_B / V_A)$

$W_{CD} = (-1/2) \times 31.0 = -15.5$ J by comparison with (b)(iii).

Net work done by gas in one cycle = $W = W_{AB} + W_{CD}$

$W = 31.0 - 15.5 = +15.5$ J [1]

Hence, efficiency = (net work done per cycle) / (heat absorbed per cycle)

= 15.5 / 45.4

= 0.341 = **34.1 %** [1]

Q4(a)